

Rainfall Estimation

This note summarizes results for a collection of 100 patches. All aggregate figures and tables that follow should therefore be read as patch-level summaries over this hundred-patch evaluation set, rather than as statistics from a single patch or from the full archive of available patches.

Relative Rainy-Pixel Distance Profile

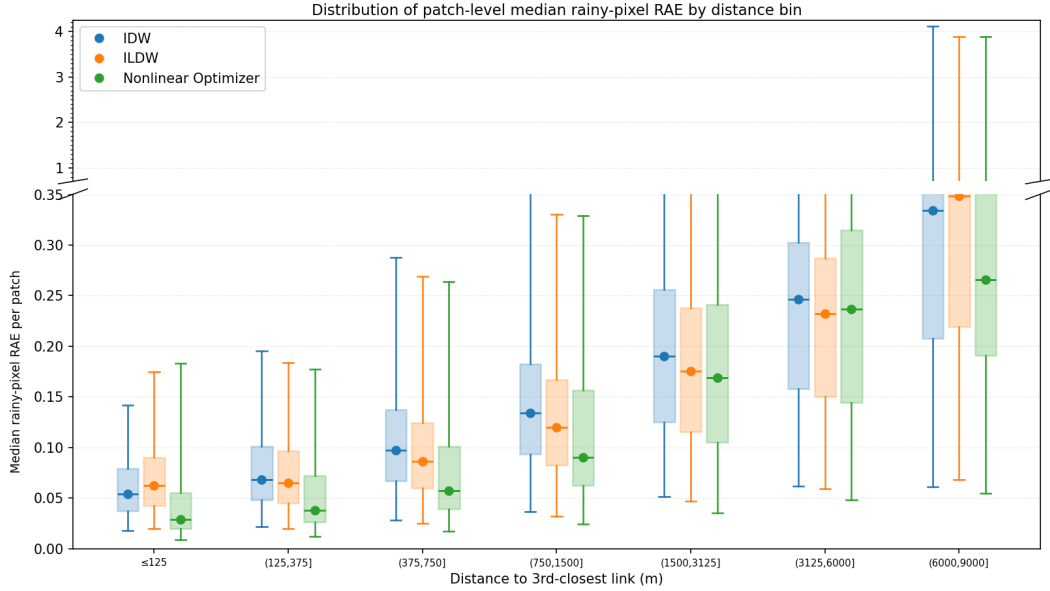


Figure 1: Rainy-pixel distance-profile box-and-whisker summary for distance to the third closest link. Each patch contributes one median rainy-pixel relative-absolute-error (RAE) value per distance bin, and these patch-level medians are then summarized across patches with a box-and-whisker plot. Here, a rainy pixel is a pixel whose ground-truth rainfall is at least the configured wet threshold (set to 0.6), and relative absolute error for a rainy pixel p is defined as $|GT(p) - PRED(p)|/GT(p)$.

This plot summarizes how solver performance changes as pixels get farther from the third-nearest link. Lower values indicate lower rainy-pixel relative absolute error in that distance bin.

Rainy-pixel Count Table for Distance to the 3rd Closest Link

The table below gives the average number of rainy pixels per patch and the corresponding standard deviation for each distance bin to the 3rd closest link. These are the same patch-level counts summarized in the tick labels of the rainy-pixel distance-profile plots.

Distance to 3rd-closest link (m)	mean(# rainy pixels)	SD
[0, 125]	4267.0	960.4
(125, 375]	16 737.7	3803.2
(375, 750]	26 809.9	6707.0
(750, 1500]	51 988.5	14 468.0
(1500, 3125]	88 839.4	27 531.1
(3125, 6000]	45 929.4	15 679.4
(6000, 9000]	3835.8	2725.7
(9000, ∞)	514.4	966.9

Table 1: Distribution of the number of pixels per distance bin. In each patch, pixels are classified into bins according to their distance to the 3rd-closest link. Formally, a pixel p belongs to bin $(d_1, d_2]$ if the disc of radius d_2 centered at p intersects at least 3 links and the disc of radius d_1 centered at p intersects at most 2 links. Let $|\text{bin}_i(P)|$ denote the number of pixels in patch P that belong to bin_i . The table reports, for each bin, the mean and standard deviation of $|\text{bin}_i(P)|$ across patches. The last bin, $(9000, \infty)$ is not shown in ?? because it is only sparsely populated.

Confusion Map

IDW			ILDW			Nonlinear Optimizer		
GT	Prediction		GT	Prediction		GT	Prediction	
	Wet	Dry		Wet	Dry		Wet	Dry
Wet	0.893 ± 0.013	0.002 ± 0.000	Wet	0.892 ± 0.013	0.003 ± 0.000	Wet	0.890 ± 0.013	0.005 ± 0.001
Dry	0.040 ± 0.004	0.065 ± 0.010	Dry	0.037 ± 0.004	0.068 ± 0.010	Dry	0.030 ± 0.004	0.075 ± 0.010

Table 2: Patch-averaged wet/dry confusion matrices for IDW, ILDW, and the nonlinear optimizer at a threshold of 0.6 mm/h, where a pixel is called wet if its rainfall is at least 0.6 mm/h and dry otherwise. For each patch, pixels are first assigned to the four confusion categories defined by the ground-truth and predicted wet/dry labels: top-left is TP, top-right is FN, bottom-left is FP, and bottom-right is TN. The count in each category is then divided by the total number of pixels in that patch, producing a patch-level pixel fraction. The table entries report the mean of those patch-level fractions across the 100 evaluated patches, together with the standard error of the mean (SEM). These are therefore fractions of all pixels in a patch, not row-normalized conditional rates.

Patch Geometry Overview

Approx. patch size (km)	# links	1-multiplicity	2-multiplicity	3-multiplicity
50×50	939	99	840	0
62×60	1254	128	1126	0
72×70	1507	157	1350	0

Table 3: Representative link multiplicities for three HundredPatches cases whose dimensions are close to 50×50 , 60×60 , and 70×70 km. Here, multiplicity is defined by grouping links whose corresponding endpoints match within 1 m in Euclidean distance, allowing endpoint order reversal. A 1-multiplicity link appears alone, a 2-multiplicity link belongs to an exactly duplicated pair, and a 3-multiplicity link belongs to a triplet of coincident links. The entries in the table count links, not multiplicity groups.